Report: Two-Switch, Two-VLAN Network and Why Inter-VLAN Traffic Needs a Router

Tool: Cisco Packet Tracer | **Topic:** VLAN segmentation, trunking, and inter-VLAN communication

1. Objective

To build a topology with two switches and two VLANs, show that hosts in the same VLAN communicate across both switches, and explain, with evidence, why hosts in different VLANs cannot communicate without a router, even when they are plugged into the same physical switch.

2. Topology

Two 2960 switches are connected by a link between them. Switch1 has PC1 and PC2; Switch2 has PC3 and PC4. The inter-switch link is configured as a trunk so both VLANs span both switches.

```
PC3   PC4                    PC1   PC2
  \\   |                       |   /
    Switch2 ===== trunk ===== Switch1
```

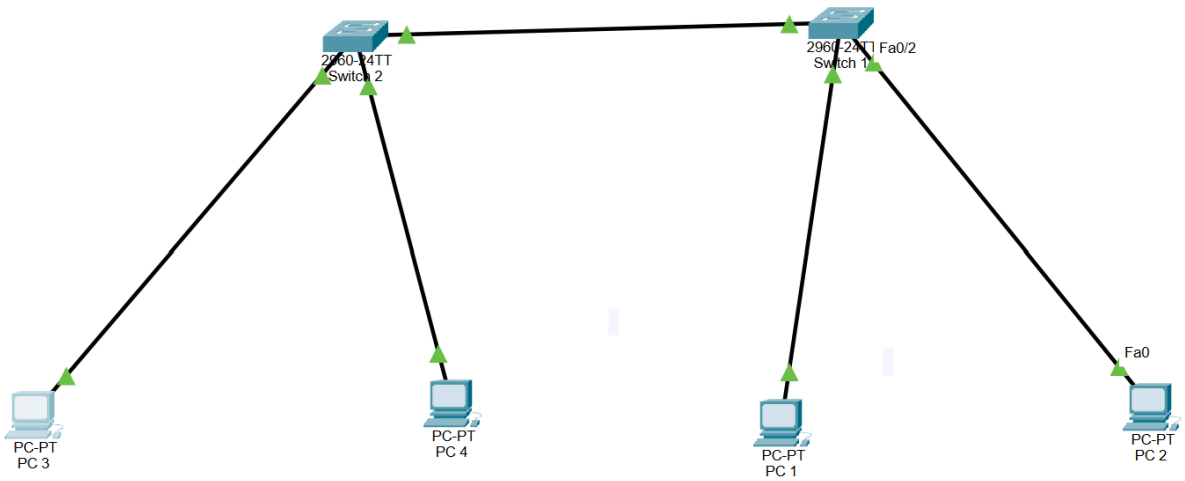


Figure 1: Packet Tracer topology with two switches and four PCs. There is no router in the network.

3. VLAN and Addressing Plan

The screenshots confirm two subnets, 192.168.10.0/24 and 192.168.20.0/24, one per VLAN. The PC-to-VLAN mapping below is the layout this report assumes; check it against each PC's Config tab and edit if yours differs.

VLAN	Subnet	Hosts (assumed)	Spans
VLAN 10	192.168.10.0/24	PC1 (192.168.10.10), PC3 (192.168.10.11)	Switch1 and Switch2
VLAN 20	192.168.20.0/24	PC2 (192.168.20.10), PC4 (192.168.20.11)	Switch1 and Switch2

All PCs use mask 255.255.255.0. No default gateway exists because there is no router.

4. Configuration

Port numbers are examples; use the ports you actually cabled.

Both switches: create the VLANs

```
enable
configure terminal
vlan 10
 name VLAN10
vlan 20
 name VLAN20
exit
```

Switch1: access ports and trunk

```
interface fa0/2
 switchport mode access
 switchport access vlan 20
interface fa0/3
 switchport mode access
 switchport access vlan 10
interface fa0/1
 switchport mode trunk
```

Switch2: access ports and trunk

```
interface fa0/2
 switchport mode access
 switchport access vlan 10
interface fa0/3
 switchport mode access
 switchport access vlan 20
interface fa0/1
 switchport mode trunk
```

An access port belongs to exactly one VLAN and sends untagged frames. A trunk port carries frames from many VLANs between switches, each tagged with its VLAN ID (IEEE 802.1Q).

5. Test Results

5.1 Same VLAN across the trunk: works

PC1 pinged 192.168.10.11 and received all 4 replies (0% loss, average 2 ms). The reply TTL is 128, which is the unchanged starting value of the Windows-style host. No router decremented it, which proves the traffic was switched at Layer 2 only: PC1 → Switch1 → trunk → Switch2 → destination.

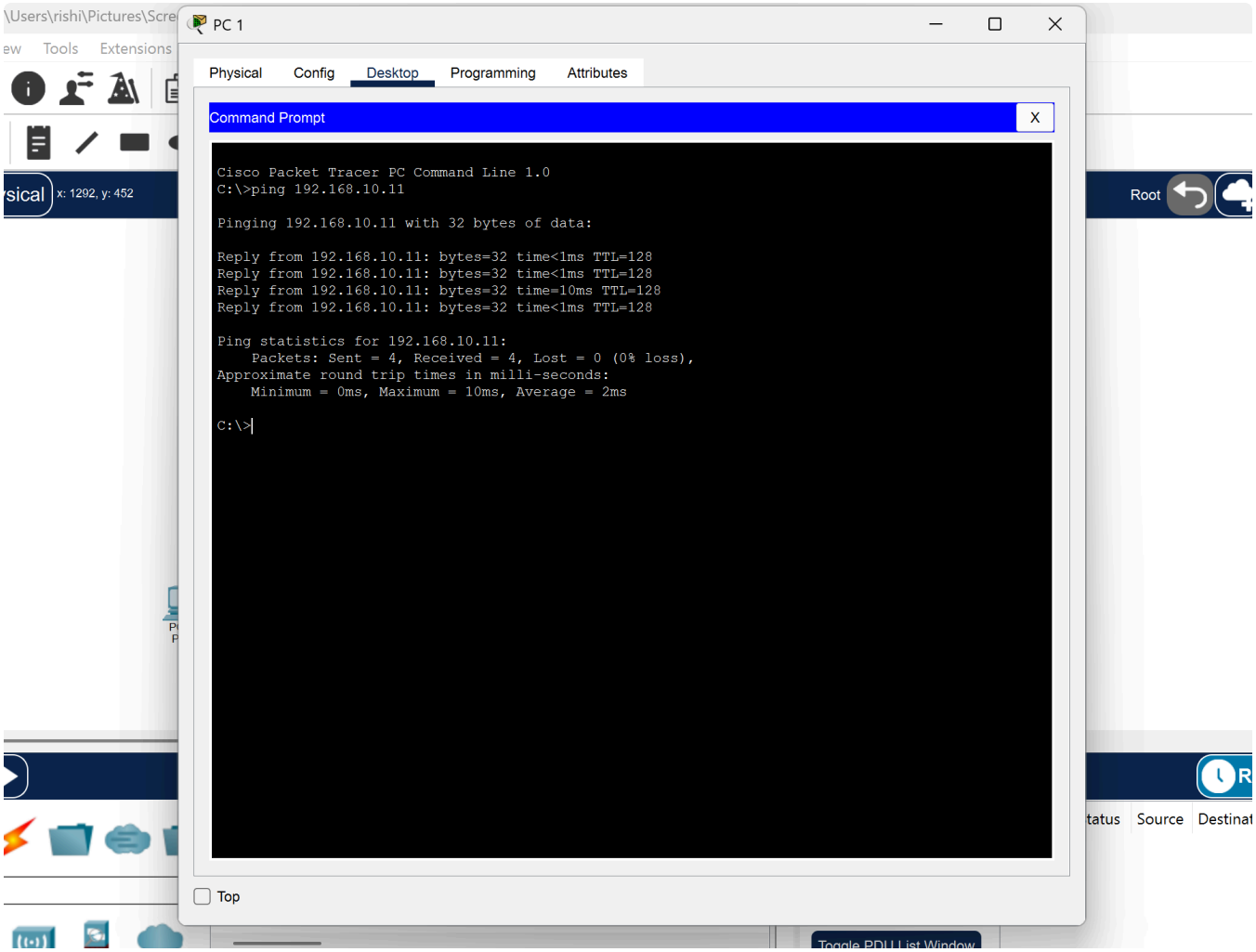


Figure 2: PC1 pinging 192.168.10.11 in the same VLAN, 0% loss, TTL 128.

5.2 Different VLAN: fails

PC3 pinged 192.168.20.11, a host in the other VLAN, and all 4 requests timed out (100% loss). The destination is reachable at the physical level (it is on the same switch or reached over the same trunk), yet the ping fails.

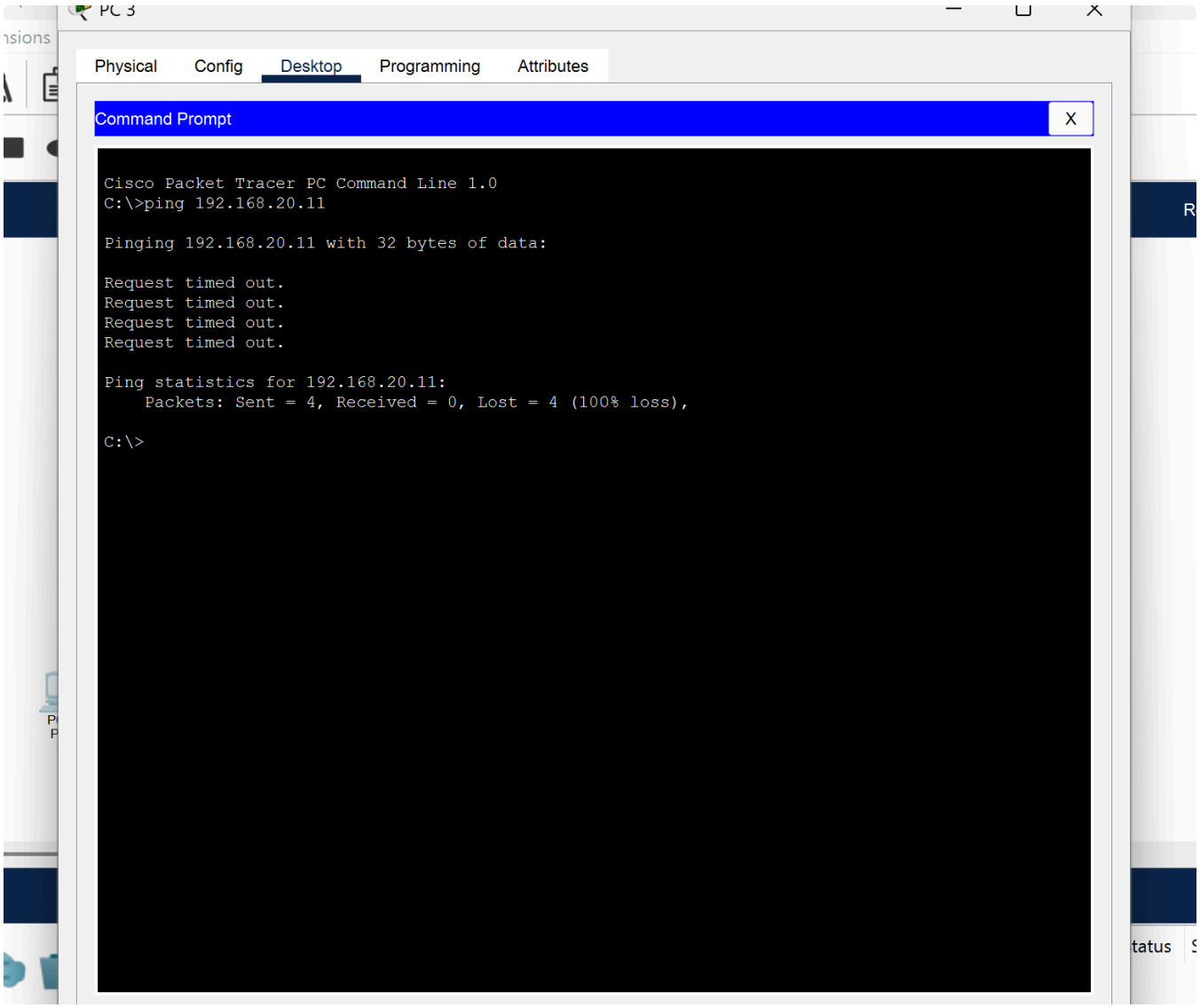


Figure 3: PC3 pinging 192.168.20.11 in a different VLAN, 100% loss.

Test	Source to destination	Same VLAN?	Result
1	PC1 to 192.168.10.11	Yes	4 of 4 replies
2	PC3 to 192.168.20.11	No	0 of 4 replies

6. Why Different VLANs Cannot Talk Without a Router

1. A VLAN is a separate broadcast domain. Each VLAN behaves like its own independent switch. A switch only forwards a frame to ports that belong to the same VLAN as the port it arrived on, and it keeps a separate MAC address table per VLAN. A frame from VLAN 10 is never delivered out of a VLAN 20 port, even when both ports are on the same chassis.

2. ARP and other broadcasts stop at the VLAN boundary. To reach 192.168.20.11, a host needs its MAC address. ARP is a broadcast, and the switch floods it only inside the sender's own VLAN. The host in the other VLAN never hears the request, so it never answers.

3. Each VLAN is a different IP subnet. The source host compares the destination address with its own subnet and sees that 192.168.20.11 is not local. For a non-local address a host does not ARP for the destination; it sends the packet to its default gateway. In this lab there is no gateway, so the packet has nowhere to go.

4. A switch works at Layer 2. A 2960 switch forwards by MAC address and VLAN. Moving a packet from one subnet to another is a Layer 3 function, which needs a device that reads IP addresses, has an interface in each subnet, and holds a routing table. That device is a router (or a Layer 3 switch).

5. The trunk does not merge VLANs. The link between the switches carries both VLANs at once, but it keeps them separate by tagging every frame with its VLAN ID. It extends each VLAN across switches; it does not connect one VLAN to another.

Putting it together: even though PC3 and its target are physically close, the switch isolates them at Layer 2, ARP cannot cross, and the source has no gateway to hand the packet to. This is what the 100% loss in Figure 3 shows, and it is the intended behavior, since it is how VLANs provide separation and security.

7. How a Router Would Fix It

This was not implemented in this lab. The usual solution is router-on-a-stick: one router link to a switch trunk port, with one sub-interface per VLAN acting as that VLAN's default gateway.

```
interface g0/0
 no shutdown
interface g0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
interface g0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
```

The PCs would then use 192.168.10.1 and 192.168.20.1 as their default gateways. A VLAN 10 host sends the packet to its gateway, the router routes it to the other sub-interface, and the packet returns to the switch tagged for VLAN 20. A Layer 3 switch with SVIs (`interface vlan 10`) can do the same job.

8. Conclusion

Hosts in the same VLAN communicated across both switches through the trunk with no router, while hosts in different VLANs could not communicate at all. VLANs split one physical switch into separate Layer 2 networks, each with its own broadcast domain and IP subnet, so crossing between them requires Layer 3 routing.

9. Screenshots Still Needed for a Complete Submission

1. `show vlan brief` on both switches, showing which ports belong to VLAN 10 and VLAN 20
2. `show interfaces trunk` on a switch, showing the trunk port and allowed VLANs
3. IP configuration of each PC (Desktop > IP Configuration) confirming addresses and VLAN mapping
4. A same-VLAN ping that stays on one switch, to complete the comparison
5. `show running-config` on both switches